MTOS

METABOLIC TZOLKIN OPERATING SYSTEM

A COMPUTATIONAL MODEL OF
ATTENTION DYNAMICS
IN A 260-DAY COGNITIVE CYCLE

ATTENTION • CYCLICITY • RESONANCE • EVOLUTION

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Metabolic Tzolkin Operating System (MTOS)

A Computational Model of Attention Dynamics in a 260-Day Cognitive Cycle

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Abstract

The Metabolic Tzolkin Operating System (MTOS) is a computational framework designed to simulate attention dynamics within a cyclic temporal structure inspired by the 260-day Tzolkin system. The model treats attention as a dynamic variable influenced by archetypal resonance, oscillatory tone phases, fatigue accumulation, and collective field effects.

The system integrates several layers of interaction:

- archetypal resonance between cognitive agents,
- oscillatory modulation based on a 13-phase tone cycle,
- fatigue-regulated attention metabolism,
- collective attention field feedback,
- adaptive learning from historical attention data,
- dynamic social interaction networks influenced by temporal conditions.

MTOS produces simulations of cognitive states, attention trajectories, attractor landscapes, and collective behavioral patterns. The system also computes metrics such as entropy, chaos indicators, Lyapunov approximations, and predictability horizons to characterize system stability.

The goal of the framework is not predictive determinism but the exploration of emergent patterns in cyclic cognitive environments.

In this sense, the MTOS framework represents an attempt to formalize the Tzolkin cycle as a **computational dynamical system**, rather than a purely calendrical structure.

Conceptual Background

Many natural and social systems evolve within cyclic temporal structures. Biological rhythms, ecological cycles, and cognitive performance patterns often display periodic behaviors that influence system dynamics.

MTOS explores the hypothesis that cognitive attention can be modeled as a metabolically regulated signal evolving within a cyclic temporal architecture. In this framework, attention is treated as a bounded dynamic variable influenced by several interacting forces:

1. Archetypal resonance
2. Oscillatory temporal phases
3. Fatigue accumulation
4. Collective field effects
5. Adaptive feedback from historical behavior

The temporal scaffold used in the model consists of a 260-step cycle formed by the interaction of two periodic sequences:

- a 20-element archetypal sequence (“seals”)
- a 13-phase oscillatory sequence (“tones”)

This structure creates a repeating lattice of cognitive conditions across time.

Within the MTOS framework, each day corresponds to a specific coordinate in this 13×20 phase space, producing a continuously evolving cognitive environment.

Temporal Reference System

The MTOS model uses a 260-day cyclic temporal structure inspired by the Mesoamerican Tzolkin system.

It is important to note that several modern interpretations of the 260-day cycle exist. One widely known contemporary system is the **Dreamspell calendar**, introduced by José Argüelles, which reinterprets the traditional structure using a fixed synchronization date (26 July 1987).

The MTOS framework adopts a **computational abstraction of the 13 × 20 cyclic structure**, rather than attempting to reconstruct the historical Mayan calendrical system with full archaeological accuracy.

In this implementation:

- the 20 archetypal positions are represented as seals
- the 13 oscillatory phases are represented as tones
- the cycle evolves deterministically through the 260 combinations

The synchronization constant used in the simulation corresponds to a Dreamspell-style epoch reference. However, within MTOS this reference serves only as a **computational coordinate system** for indexing the 260-phase lattice.

The purpose of the model is therefore not calendrical reconstruction but **the exploration of cyclic cognitive dynamics within a structured temporal phase space.**

Cybernetic Interpretation

The MTOS framework can also be interpreted within the tradition of cybernetics and complex systems theory.

Classical cybernetics, as developed by Norbert Wiener and W. Ross Ashby, focused on feedback regulation and adaptive behavior in dynamic systems. In these frameworks, system stability emerges through continuous interaction between internal states and environmental signals.

MTOS extends this perspective by introducing a **cyclic temporal environment** as an explicit structural component of the system. Rather than treating time as a neutral parameter, the model assumes that the environment itself evolves through a structured sequence of cognitive conditions.

Within this perspective, the 260-phase temporal lattice functions as a **dynamic cognitive environment**, while attention acts as the regulated variable responding to resonance, oscillation, fatigue, and collective feedback.

In this sense, MTOS may be viewed as an exploratory example of **cybernetic modeling within cyclic temporal architectures**, where cognition is studied as a dynamic process emerging from the interaction between agents and a structured temporal field.

Ecological Interpretation of Attention Dynamics

Another useful perspective on the MTOS framework is its similarity to ecological system models.

In ecology, complex systems are often described as networks of interacting agents exchanging energy within a structured environment. System behavior emerges from feedback between resource availability, metabolic constraints, and environmental conditions.

The MTOS model exhibits several structural similarities with such systems.

In this framework:

- attention functions as a bounded energetic resource
- fatigue represents metabolic limitation
- the tone cycle introduces seasonal oscillation in system dynamics

- archetypal resonance models interaction energy between agents and environmental conditions
- the global attention field acts as a slowly evolving environmental memory

From this viewpoint, MTOS can be interpreted as a **minimal ecological model of cognitive energy dynamics** evolving within a cyclic temporal habitat.

This interpretation helps explain the emergence of attractors, oscillatory regimes, and collective attention climates observed in simulations.

Rather than describing individual cognition directly, the system models the **environmental conditions under which cognitive states evolve**.

System Architecture

The MTOS model is composed of several interacting subsystems that collectively generate the dynamics of attention.

The architecture contains the following major components.

Figure 1 — MTOS System Architecture

The MTOS system is composed of several interacting modules forming a cyclic computational pipeline.

```
+-------------------------+
|   Temporal Engine  |
|    (260 day cycle)   |
+-----------+------------+
            |
            v
+-------------------------+
|      Archetypal        |
| Resonance Field    |
+-----------+-------------+
            |
            v
```

```
+-------------------------+
|    Tone Oscillator    |
| (13 phase wave)    |
+-----------+-------------+
            |
            v
+-------------------------+
| Attention Dynamics |
|   + Fatigue Model    |
+-----------+-------------+
            |
            v
+------------------------+
| Global Attention   |
|        Field             |
+-----------+-----------+
            |
            v
+------------------------+
| Adaptive Learning  |
+-----------+-------------+
            |
            v
+-------------------------+
| Simulation Output  |
| Forecast / Metrics |
```

+----------+

Figure 1 illustrates the conceptual architecture of the Metabolic Tzolkin Operating System (MTOS).

The system integrates a temporal engine, archetypal resonance field, tone oscillator, metabolic attention dynamics, global attention field, and adaptive learning module.

These components interact iteratively to generate attention trajectories and collective cognitive climate patterns.

1. Temporal Engine

The temporal engine maps calendar dates to positions within the 260-step cycle.

Each day is represented by:

- a seal index (0 – 19)
- a tone index (1 – 13)

This pair defines the archetypal and oscillatory properties of the day.

1.1 Phase Space Representation

The MTOS temporal structure can be interpreted as a discrete phase space formed by the Cartesian interaction of two periodic dimensions:

- a 20-state archetypal axis (seals)
- a 13-phase oscillatory axis (tones)

This produces a **13 × 20 lattice** containing 260 unique states.

Each day within the simulation corresponds to a coordinate within this lattice. As time advances, the system moves deterministically through the phase space, generating a structured sequence of cognitive environments.

This interpretation allows the MTOS cycle to be treated as a **finite dynamical system operating within a discrete temporal phase space.**

2. Archetypal Resonance Field

Agents possess their own archetypal identity defined by a seal and tone. Interaction between archetypes produces resonance values based on geometric relationships within the seal sequence.

Key resonance relations include:

- identity resonance

- analog resonance
- occult resonance
- antipode opposition

These relationships generate positive or negative influence on attention dynamics.

3. Oscillatory Tone Modulation

The 13-phase tone sequence introduces a cyclic oscillation into the system. Tone influence is modeled as a cosine wave affecting attention intensity.

This creates periodic amplification and attenuation of cognitive energy within the cycle.

4. Attention Metabolism

Interpretation of Attention as a Metabolic Variable

In the MTOS framework, attention is treated as a metabolically regulated signal rather than a static cognitive resource.

This interpretation draws inspiration from biological systems where energy allocation, fatigue accumulation, and recovery processes regulate system activity.

Attention in MTOS therefore behaves similarly to a **metabolic variable**, subject to both amplification (resonance and oscillatory phase alignment) and suppression (fatigue accumulation).

This approach allows the system to produce realistic bounded dynamics rather than unlimited growth of cognitive activation.

Attention is modeled as a bounded variable evolving through iterative updates. Each update step incorporates:

- archetypal resonance with the current day
- tone-based oscillatory modulation
- fatigue accumulation
- fatigue decay

Fatigue represents a metabolic constraint on attention and prevents runaway growth of the variable.

5. Collective Attention Field

MTOS also maintains a global field representing accumulated attention levels across the 260-day cycle.

This field evolves through a slow learning process:

- each simulated day contributes a fraction of its attention value
- the field gradually stabilizes into a distribution of collective cognitive attractors

The field then feeds back into individual simulations, creating collective influence.

6. Adaptive Learning

Historical attention data stored in the system modifies initial conditions of simulations.

Two learning processes operate simultaneously:

- trend learning from recent attention trajectories
- volatility-based adjustment from longer historical windows

These adjustments simulate adaptive behavior within the cognitive system.

Mathematical Model

MTOS models attention as a bounded dynamic variable evolving under the influence of archetypal resonance, oscillatory modulation, fatigue dynamics, and collective field feedback.

Let:

- a_t — attention level at time step t
- f_t — fatigue level at time step t
- r — archetypal resonance between the agent and the current day
- ω — tone oscillation effect
- F — collective field influence

Attention is updated iteratively according to the following process.

Archetypal Resonance

Resonance between two archetypes is defined as a discrete mapping within the 20-element seal space:

$$r(a,b) = \begin{cases} 0.30 & \text{same seal} \\ 0.18 & \text{analog relation} \\ 0.12 & \text{occult relation} \\ -0.28 & \text{antipode relation} \\ 0 & \text{otherwise} \end{cases}$$

This mapping defines the basic interaction energy between archetypal identities.

Tone Oscillation

The influence of the tone cycle is modeled as a periodic oscillator:

$$\omega(t) = 0.06 \cos\left(\frac{2\pi \cdot tone}{13}\right)$$

This introduces periodic fluctuations in attention across the 13-phase cycle.

Fatigue Dynamics

Fatigue acts as a metabolic regulator preventing unlimited attention growth.

The fatigue update rule is:

$$f_{t+1} = f_t + 0.05a_t - 0.03$$

with clipping to the interval:

$$0 \leq f \leq 1$$

Fatigue then suppresses attention:

$$a_t = a_t - 0.07 f_t$$

Energetic Interpretation of Attention Regulation

The regulation of attention in MTOS can also be interpreted through the lens of energetic optimization.

In biological and cognitive systems, activity is often constrained by metabolic costs and regulatory feedback mechanisms. These constraints prevent uncontrolled growth of internal signals and promote stable system behavior.

Within the MTOS framework, the fatigue variable introduces a metabolic constraint that continuously suppresses excessive attention activation. At the same time, resonance and oscillatory tone alignment amplify attention when environmental conditions are favorable.

This interaction produces a dynamic balance between amplification and suppression.

Conceptually, such dynamics resemble regulatory principles found in theoretical neuroscience, including the idea that cognitive systems tend to stabilize internal activity within bounded energetic regimes.

Although MTOS does not explicitly implement variational inference or probabilistic prediction, the system demonstrates how **simple feedback constraints can produce stable regimes of cognitive activity within structured temporal environments**.

Collective Field Feedback

MTOS maintains a global attention field across the full 260-day cycle.

Let:

Gk

represent the average collective attention for Kin k.

The field is updated through exponential smoothing:

$$G_k = 0.9G_k + 0.1a_t$$

Individual attention dynamics are influenced by this field through:

$$a_t = a_t + 0.2(G_k - 0.5)$$

This mechanism introduces collective memory into the system.

Simulation Process

The MTOS simulation iterates attention dynamics across a selected time horizon.

Each simulation step includes:

1. Determining the Kin index of the current day.
2. Computing archetypal resonance between the user and the day.
3. Applying tone oscillation.
4. Updating fatigue.
5. Applying fatigue suppression.
6. Incorporating collective field influence.
7. Storing the resulting attention value.

The resulting attention series represents a trajectory through the cognitive climate of the 260-day cycle.

Attention State Classification

Attention values are mapped to qualitative cognitive states:

Attention Level	Cognitive State
0.80	Focus
0.65–0.80	Flow
0.45–0.65	Neutral
0.30–0.45	Fatigue
< 0.30	Recovery

This classification allows MTOS to interpret numeric trajectories in human-readable form.

Attractor Analysis

To explore stable patterns within the cognitive system, MTOS computes a resonance attractor map.

For every pair of archetypes (s1, s2), the system simulates multiple iterations of attention dynamics under all combinations of tone phases.

The resulting matrix represents average long-term attention convergence between archetypes.

This produces a 20×20 attractor landscape identifying:

- stable cognitive partnerships
- unstable or conflicting archetype combinations
- neutral interaction regions.

System Metrics

Several quantitative metrics are computed to characterize the behavior of attention trajectories.

Entropy

Entropy measures the distribution of attention states:

$$H = -\sum p_i \log(p_i)$$

Higher entropy indicates greater variability in attention.

Chaos Indicator

Chaos is approximated using the standard deviation of successive attention differences:

$$C = std(\Delta a_t)$$

This captures the degree of short-term instability.

Lyapunov Estimate

A simplified Lyapunov estimate measures divergence in attention dynamics:

$$L = mean(\log(|\Delta a_t|))$$

Positive values indicate sensitive dependence on initial conditions.

Predictability Horizon

Predictability measures how long the trajectory remains stable before large deviations occur.

The horizon is defined as the first time step where:

$$|\Delta a_t| > 0.12$$

This provides an approximate predictability limit of the system.

Collective Attention Model

MTOS aggregates attention values from all users stored in the database.

The collective model computes:

- mean collective attention
- volatility of collective attention

These statistics represent the global cognitive climate of the system.

Dynamic Social Interaction Network

MTOS also models interactions between agents.

Each pair of agents is assigned a dynamic interaction score based on:

1. Archetypal resonance between seals
2. Tone compatibility between individuals
3. Resonance of each agent with the current day
4. Tone resonance with the day's tone

The resulting score produces interaction categories:

Score	Interaction Type
0.25	Strong
0.10–0.25	Collaborate
-0.10–0.10	Neutral
-0.25–-0.10	Ignore
< -0.25	Conflict

Because the day changes continuously, the interaction network evolves over time.

Computational Implementation

The MTOS framework is implemented in Python using a modular architecture designed to allow reproducible simulations and extensibility.

The core implementation relies on the following components:

- `numpy` for numerical computations
- `datetime` for temporal mapping of calendar dates to positions within the 260-day cycle
- `json` for persistent storage of users, attention history, and system metrics
- `os` for file system interaction

A deterministic random seed is used to ensure reproducibility of simulations:

np.random.seed(42)

The system stores persistent data in three main structures:

1. **User Database**

mtos_users.json

Stores user archetypes (seal and tone) and birth information.

2. **Attention History**

mtos_attention_db.json

Stores daily attention values for each user.

3. **Global Field Memory**

mtos_global_field.json

Stores collective attention values across the 260-day cycle.

Simulation Pipeline

A typical MTOS simulation follows this sequence:

1. Register or load a user archetype.
2. Determine the Kin index for the current day.
3. Initialize attention and fatigue levels.
4. Run a forward simulation across a chosen horizon (e.g., 7, 30, or 260 days).
5. Store the resulting attention trajectory.
6. Update the global attention field.
7. Compute system metrics.
8. Generate forecasts and cognitive state windows.
9. Evaluate social interaction networks.

This pipeline allows the system to evolve as more data accumulates.

Experimental Observations

Although MTOS is primarily designed as an exploratory computational model, several recurring behaviors appear in simulations.

1. Attention Oscillation

Attention trajectories tend to display quasi-periodic oscillations driven by the 13-tone cycle.

These oscillations are modulated by fatigue accumulation and collective field feedback.

2. Emergence of Cognitive Windows

Simulations frequently produce contiguous windows of elevated attention values.

These "focus windows" represent periods of high cognitive stability within the cycle.

Such windows may last between 3 and 7 days in typical simulations.

3. Archetypal Attractor Structures

The attractor map often reveals stable interaction zones within the archetypal space.

Certain seal combinations consistently produce higher attention equilibrium levels, suggesting the existence of archetypal attractors in the system.

4. Social Resonance Dynamics

The dynamic social network reveals that interactions between agents vary depending on the current day within the cycle.

This produces a continuously evolving interaction topology in which collaboration potential fluctuates with the temporal environment.

Potential Applications

Although MTOS is currently an experimental framework, the underlying approach suggests several potential applications.

Cognitive Rhythm Analysis

The system may serve as a tool for exploring cyclic patterns in attention and productivity.

Collective Attention Modeling

The global attention field can simulate collective cognitive climates within groups of interacting agents.

Social Interaction Forecasting

The dynamic interaction network provides a method for studying temporal fluctuations in collaborative potential between agents.

Simulation of Cognitive Environments

MTOS can be used as a sandbox environment for exploring the dynamics of cognitive energy within structured temporal systems.

Cognitive Climate Model

The MTOS framework can also be interpreted as a simplified **climate model of cognitive dynamics**.

In climate science, large-scale environmental behavior emerges from the interaction of multiple coupled variables such as temperature, pressure, and energy flows. Similarly, MTOS treats attention as a dynamic variable evolving within a structured temporal environment.

Within this perspective:

- the 260-day cycle represents the **temporal climate system**
- seals define the **archetypal landscape of cognitive conditions**
- tones introduce **oscillatory energy phases**
- fatigue acts as a **metabolic regulator**
- the collective attention field functions as a **global climate memory**

Together these mechanisms generate a continuously evolving **cognitive climate**, within which individual agents move through phases of focus, fatigue, recovery, and flow.

Rather than predicting individual behavior, the model explores **patterns of stability, attractor formation, and collective resonance within the cognitive environment**.

This interpretation suggests that cyclic temporal systems may influence cognition in a manner analogous to environmental climate systems influencing biological activity.

Limitations

Several limitations must be acknowledged.

Simplified Archetypal Geometry

The resonance mapping between archetypes is intentionally simplified and may not capture the full complexity of symbolic systems.

Limited Empirical Calibration

The current model is exploratory and has not yet been calibrated against large-scale empirical cognitive datasets.

Deterministic Dynamics

Although stochastic processes could be incorporated, the current model is largely deterministic apart from initialization.

Small Population Effects

The collective field dynamics become more meaningful as the number of agents increases.

Small user populations may produce noisy collective metrics.

Future Work

Several directions could further extend the MTOS framework.

Large-Scale Agent Simulations

Running simulations with hundreds or thousands of agents would allow exploration of large collective attention fields.

Graph-Based Network Visualization

Visualizing the dynamic interaction network could reveal structural clusters within archetypal space.

Multi-Level Learning

Future versions may introduce hierarchical learning where both individual agents and the global field adapt over time.

Integration with External Data

Incorporating real behavioral or productivity datasets could provide empirical grounding for the model.

Conclusion

The Metabolic Tzolkin Operating System (MTOS) introduces a computational approach for exploring attention dynamics within cyclic temporal structures.

By combining archetypal resonance, oscillatory tone modulation, fatigue regulation, adaptive learning, and collective feedback, the system produces a rich dynamic landscape of cognitive behavior.

Rather than serving as a predictive tool, MTOS functions as a conceptual simulation environment for investigating how cyclic temporal architectures may shape attention and social interaction patterns.

The framework demonstrates how symbolic temporal systems can be translated into computational models capable of generating measurable dynamics, attractor landscapes, and evolving interaction networks.

This work demonstrates how symbolic temporal systems can be transformed into computational models capable of generating measurable cognitive dynamics.

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